3

MEC 424 - PIPING DESIGN LECTURE NOTES

Types of joints; solder or wiped, sweet, slipe, bell and spigot, threaded or screwed joints

Different types of joints provide various options for connecting pipes and fittings, offering flexibility, strength, and ease of installation in plumbing and piping systems. The specific joint used depends on factors such as the type of material, system requirements, intended application, and industry standards. Some of these joints include;

1. Solder or Wiped Joints:
    - Solder joints, also known as wiped joints, are commonly used to join copper pipes.
    - The joint is created by applying heat to the copper fittings and pipe, melting solder, and allowing it to flow into the gap between the fittings and pipe.
    - As the solder cools and solidifies, it forms a strong and watertight connection.
2. Sweat Joints:
    - Sweat joints are a type of solder joint used specifically for copper pipes.
    - The joint is created by heating the fittings and pipe using a torch, and then applying solder to the joint, which melts and flows into the gap between the fittings and pipe.
    - The name "sweat" comes from the process of heating the copper to the point where it appears to sweat before applying solder.
3. Slip Joints:
    - Slip joints, also known as compression joints, are commonly used in plumbing fixtures such as faucets and traps.
    - These joints involve a compression fitting that allows for easy assembly and disassembly without the need for soldering or welding.
    - The joint is formed by sliding the compression nut onto the pipe, followed by a compression ring or ferrule. The nut is then tightened, compressing the ring onto the pipe, creating a watertight seal.
4. Bell and Spigot Joints:
    - Bell and spigot joints are used in joining pipes that have a bell-shaped end and a corresponding spigot end.
    - The bell-shaped end of one pipe fits over the spigot end of another, creating a mechanical connection.

- These joints are commonly found in plastic pipes, such as PVC or ABS, and are often used in sewer, drainage, and underground applications.

5. Threaded or Screwed Joints:
   - Threaded joints involve pipes with threaded ends that can be screwed together to create a secure connection.
   - The joint is formed by threading the ends of the pipes and then using pipe thread sealant or tape to ensure a tight seal.
   - Threaded joints are commonly used in metal pipes and fittings, such as galvanized steel or stainless steel, and are often found in gas lines, water lines, and plumbing fixtures.

The use of each of the joints described above

Below is a breakdown of the use and application of each joint:

1. Solder or Wiped Joints:
   - Solder joints are commonly used in copper pipe systems for water supply, heating, and cooling applications.
   - They provide a secure and leak-free connection, particularly in residential and commercial plumbing systems.
   - Solder joints are widely used in potable water systems, including faucets, valves, and fittings.
2. Sweat Joints:
   - Sweat joints, also known as solder joints, are predominantly used in copper plumbing systems.
   - They are commonly found in residential and commercial water supply lines, both hot and cold.
   - Sweat joints are used in applications such as connecting copper pipes, fittings, and valves in potable water systems.
3. Slip Joints:
   - Slip joints are frequently used in plumbing fixtures that require easy assembly and disassembly, such as faucets, traps, and other drain connections.

- They provide a convenient way to connect and disconnect pipes without the need for soldering or specialized tools.
- Slip joints are commonly used in kitchen sinks, bathroom sinks, bathtubs, and other fixtures where periodic maintenance or replacement may be required.

4. Bell and Spigot Joints:
   - Bell and spigot joints are commonly used in plastic pipe systems, particularly for sewer, drainage, and underground applications.
   - They provide a simple and secure means of connecting pipes in these applications, ensuring a tight and watertight seal.
   - Bell and spigot joints are often used in municipal sewer lines, stormwater systems, and underground drainage pipes.
5. Threaded or Screwed Joints:
   - Threaded joints are widely used in various applications, including plumbing, gas lines, and industrial piping systems.
   - They provide a reliable connection that can withstand high pressures and temperatures.
   - Threaded joints are commonly found in water supply lines, natural gas lines, fire sprinkler systems, and hydraulic systems.

N/B. Each type of joint offers specific advantages based on the material being used, the application requirements, and the desired ease of assembly or disassembly. It's important to consider factors such as the type of piping material, system compatibility, pressure and temperature requirements, and local plumbing codes and standards when selecting the appropriate joint for a given plumbing or piping project.

Constructional details of each of the joints explained above

1. Solder or Wiped Joints:
   - Solder joints are created by applying heat to the copper fittings and pipe to melt the solder.
   - The joint is assembled by cleaning the surfaces of the fittings and pipe, applying flux to prevent oxidation, and then heating the joint with a torch.

- Once the joint is heated, solder wire or paste is applied to the joint, and it flows into the gap between the fittings and pipe through capillary action.
- As the joint cools, the solder solidifies and forms a strong and watertight connection.

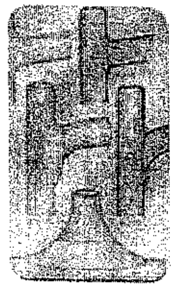
2. Sweat Joints:
    - Sweat joints are a type of solder joint used for copper pipes.
    - The process begins by cleaning the fittings and pipe surfaces, applying flux to prevent oxidation, and heating the joint using a torch.
    - The heat causes the solder to melt and flow into the gap between the fittings and pipe.
    - The joint is typically heated until the solder evenly covers the gap and forms a smooth connection.
    - As the joint cools, the solder solidifies and creates a reliable and leak-free joint.
3. Slip Joints:
    - Slip joints consist of two main components: a compression nut and a compression ring or ferrule.
    - The joint is assembled by sliding the compression nut onto the pipe, followed by the compression ring or ferrule.
    - The pipe end is then inserted into the receiving fitting or connector.
    - The compression nut is tightened onto the fitting, compressing the ring or ferrule against the pipe.
    - The compression creates a seal, preventing leaks and ensuring a secure connection.
4. Bell and Spigot Joints:
    - Bell and spigot joints are commonly used in plastic pipes, such as PVC or ABS.
    - The bell-shaped end of one pipe fits over the spigot (plain, tapered) end of another pipe.
    - The joint relies on the interference fit between the bell and spigot to create a mechanical connection.
    - The joint may be further secured with adhesive or sealant specifically designed for plastic pipes.
    - The resulting connection provides a tight seal and structural integrity for sewer, drainage, and underground applications.
5. Threaded or Screwed Joints:
    - Threaded joints involve pipes with threaded ends that can be screwed together.

PIPING JOINTS

People also ask :

1 What is a wiped joint?

The defining characteristic of a wiped joint is that the soldering process involves mechanically working or 'wiping' the joint. As well as heating solder and applying it to the joint, the solder is shaped into place manually, wiping it with a non-metallic tool to form a smooth-surfaced outer shape.



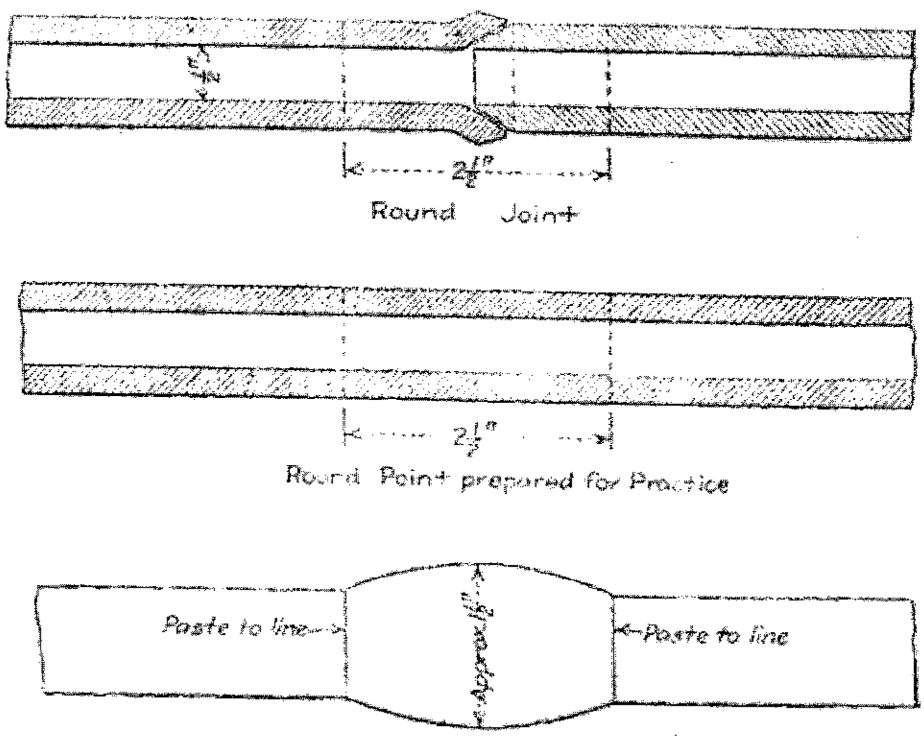


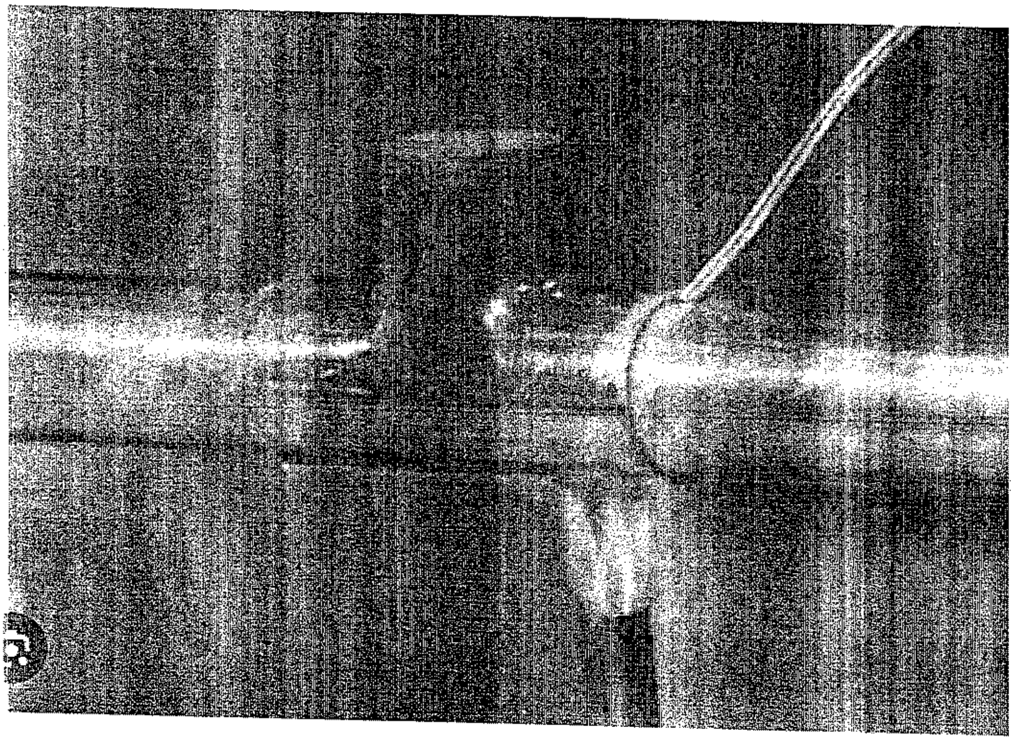
FIG. 23.

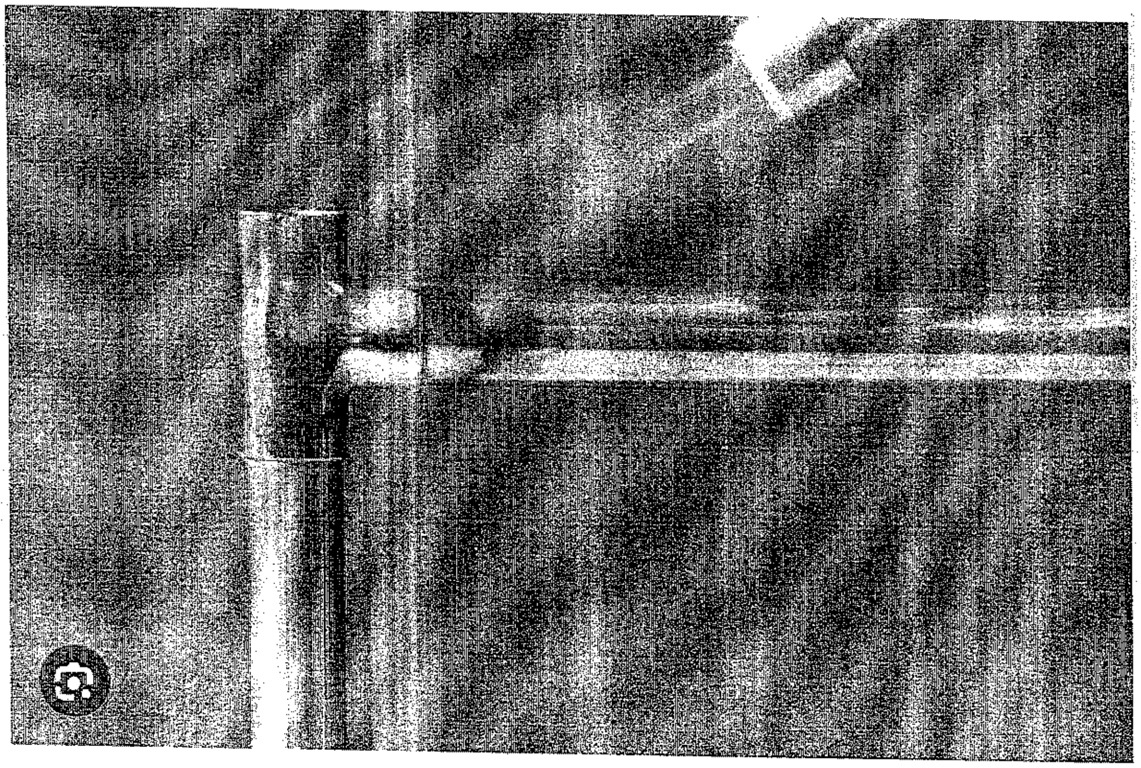
Half-Inch Round Wiped Joint

Visit

2. What is a sweat joint?

Soldering a pipe joint or "sweating a pipe" is accomplished by heating a copper fitting with a propane torch until the fitting is just hot enough to melt metal solder. The heat draws the solder into the gap between the fitting and pipe to form a water tight seal.

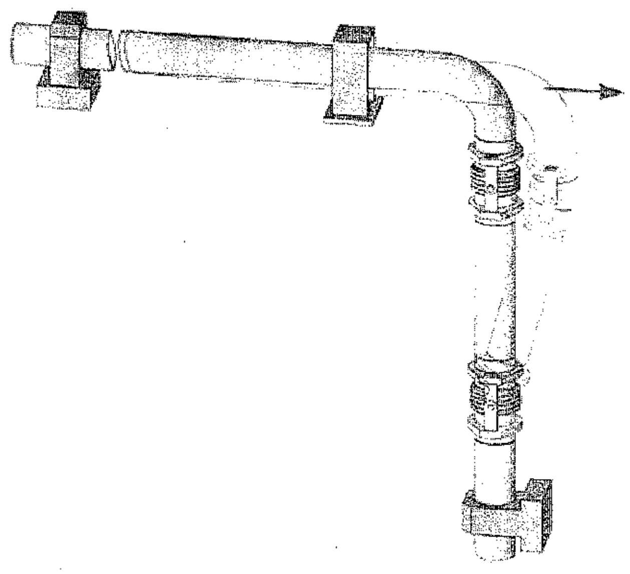




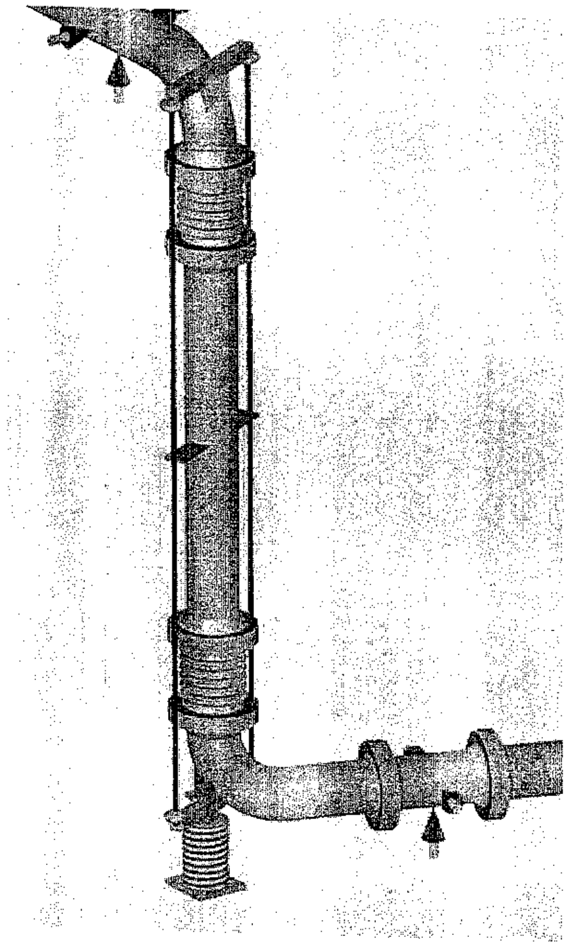
How to Sweat Copper Pipe (DIYer's Guide) - Bob Vila   Visit

3 What is a slip joint in piping?

A slip joint is a type of joint used in plumbing to connect pipes and fittings. It consists of a connecting nut and a gasket (often made of rubber or neoprene) that are placed between the end of one pipe and the fitting. The joint is then tightened with a wrench, compressing the gasket and forming a seal.



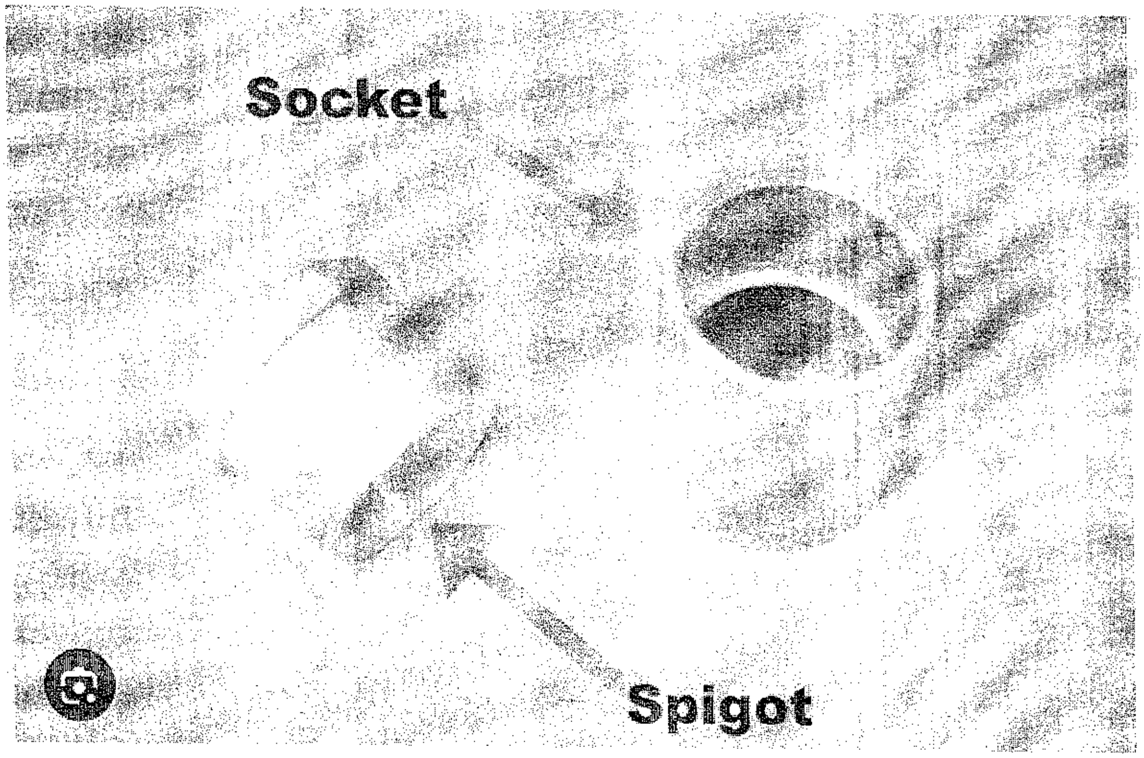
Pipe expansion bellows - Pipe expansion joints - Flexible joints for pipes - Spiroflex

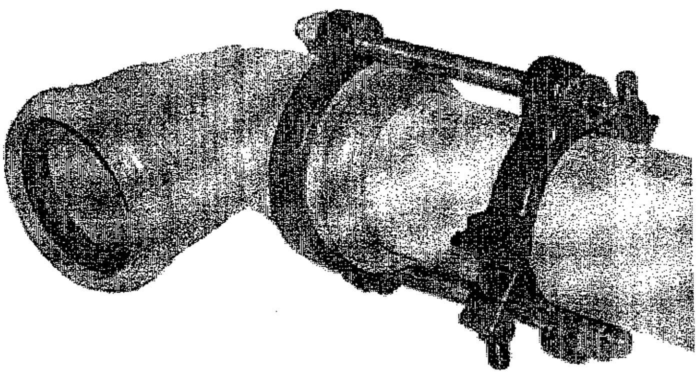


10 Expansion Joint Basics to Remember ... for Pipe Stress Analysts

4 What is bell and spigot joint?

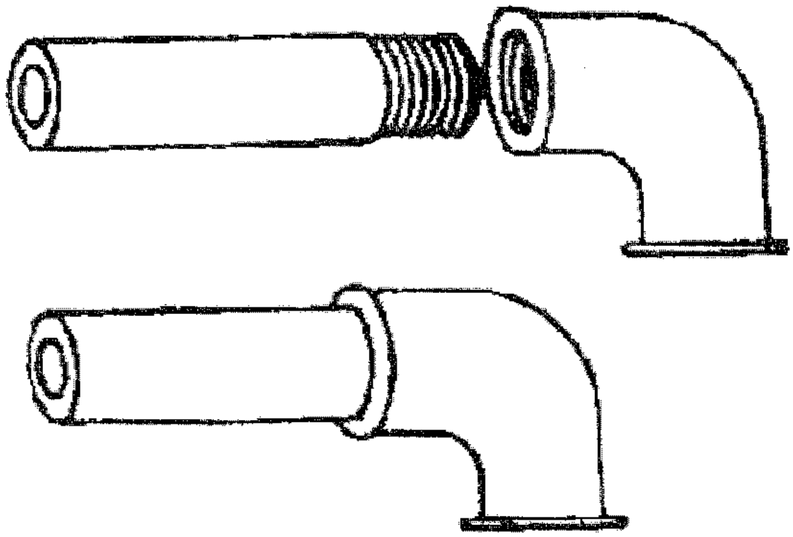
A form of joint used on pipes that have an enlarged diameter or bell at 1 end and a spigot at the other that fits into and is laid in the bell.





| Serrated Joint Restraint | Bell & Spigot PVC

5. **SCREWED OR THREADED JOINTS**



Threaded Joints - Course: Techniques of Fitting and Assembling Component Parts to Produce Simple Units....

- The joint is created by threading the ends of the pipes using a die or threading machine.
- Pipe thread sealant or tape is applied to the threads to ensure a tight seal and prevent leaks.
- The threaded ends of the pipes are then screwed together, creating a secure and durable connection.
- Threaded joints are commonly used in metal pipes, such as galvanized steel or stainless steel, and can withstand high pressures and temperatures.

Typical joining methods for given installations as well as suitable supports for the pipes

1. Water Supply Systems (Non-Pressure):
    - Joining Methods: Soldering or brazing for copper pipes, solvent welding for PVC or CPVC pipes, and compression fittings for plastic or metal pipes.
    - Suitable Supports: Plastic or metal pipe hangers, adjustable pipe clamps, and pipe straps.
2. Water Supply Systems (Pressure):
    - Joining Methods: Threaded connections with pipe thread sealant or tape for metal pipes, press fittings for copper or PEX pipes, and fusion or mechanical joints for HDPE pipes.
    - Suitable Supports: Metal pipe hangers, clevis hangers, and strut channels with pipe clamps.
3. Gas Supply Systems:
    - Joining Methods: Threaded connections with pipe thread sealant or tape for metal pipes, compression fittings for copper or stainless steel pipes, and mechanical joints for PE or PVC pipes.
    - Suitable Supports: Metal pipe hangers specifically designed for gas piping, adjustable pipe clamps, and strut channels with pipe clamps.
4. Heating and Cooling Systems:
    - Joining Methods: Soldering or brazing for copper pipes, press fittings for copper or PEX pipes, and threaded connections for steel pipes.
    - Suitable Supports: Metal pipe hangers, adjustable pipe clamps, and pipe straps.
5. Drainage and Sewer Systems:

- Joining Methods: Solvent welding or rubber gasket joints for PVC or ABS pipes, bell and spigot joints for cast iron or clay pipes, and mechanical joints for ductile iron pipes.
- Suitable Supports: Pipe supports designed for drainage systems, such as pipe clamps, pipe rollers, and hangers with resilient inserts.

6. Fire Sprinkler Systems:
    - Joining Methods: Threaded connections with pipe thread sealant for steel pipes, groove connections for ductile iron pipes, and press fittings for copper or PEX pipes.
    - Suitable Supports: Metal pipe hangers with resilient inserts, strut channels with pipe clamps, and fire-rated hangers as required by local codes.

When selecting joining methods and supports, it is crucial to refer to the specific requirements of the installation, local codes and regulations, and manufacturer guidelines. These considerations will help ensure the integrity, safety, and compliance of the pipework system. Consulting with qualified professionals, such as plumbers or mechanical engineers, is highly recommended to determine the most suitable joining methods and supports for your specific installation.

Pipe fittings such as elbows, reducers tees etc

Pipe fittings are components used to connect and redirect pipes in a plumbing or piping system. They come in various shapes, sizes, and types to accommodate different needs and configurations. Some typical examples of these fittings include;

1. Elbow: Elbows are fittings that change the direction of a pipe. They are typically available in 45-degree and 90-degree angles, allowing pipes to be bent at those angles. Elbows help navigate around obstacles or create turns in the pipe system.
2. Tee: Tees have a T-shaped design, with one inlet and two outlets (or vice versa). They allow for a branch line to be connected at a right angle to the main pipe, creating a branching connection.
3. Reducer: Reducers are fittings used to connect two pipes of different sizes. They come in two types: concentric reducers, where the ends are aligned in a straight line, and eccentric reducers, where the ends are offset.
4. Coupling: Couplings are fittings used to join two pipes of the same size. They are typically used when a permanent connection is desired. and they provide a leak-resistant joint.

5. Union: Unions are similar to couplings, but they allow for the disconnection and reconnection of pipes without requiring the fittings to be cut or disassembled. Unions are often used when periodic maintenance or repairs are anticipated.
6. Cross: Cross fittings have a plus-shaped design with four connection points. They allow for the branching of pipes in multiple directions, typically at right angles to each other.
7. Flange: Flanges are flat, circular fittings with holes around the perimeter. They are used to connect pipes by bolting them together, creating a secure and leak-resistant joint. Flanges are commonly used in high-pressure or large-diameter piping systems.
8. Cap: Caps are fittings that are used to seal the end of a pipe. They are typically used for terminating a line or temporarily closing off a pipe for maintenance or testing purposes.

These are just a few examples of pipe fittings commonly used in plumbing and piping systems. It's important to note that there are many more types of fittings available, and each serves a specific purpose in a piping system. The selection of the appropriate fittings depends on factors such as pipe size, material, fluid type, system requirements, and local codes and standards.

Uses of fittings such as elbows, reducers, tees etc

Common uses of some popular pipe fittings:

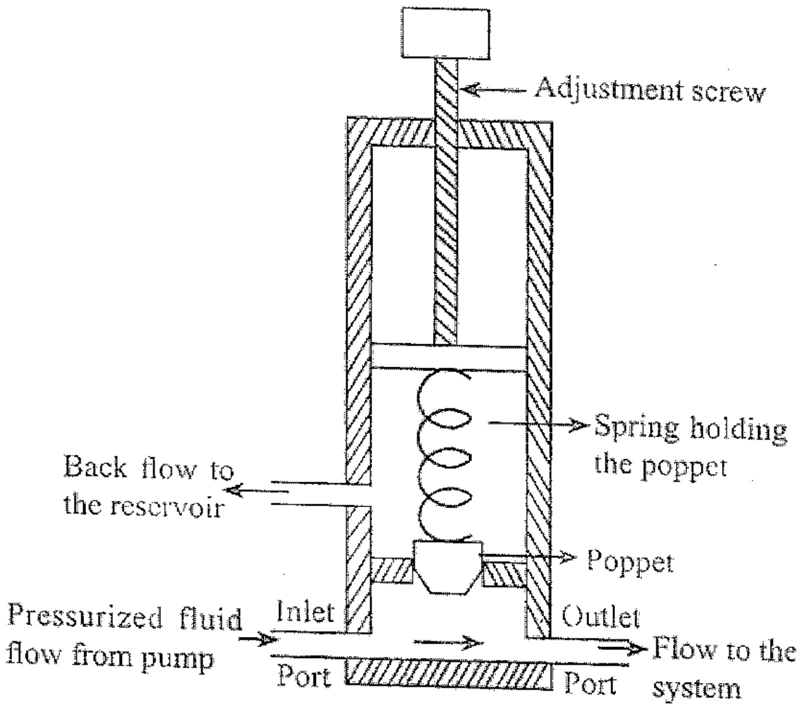
1. Elbow:
    - Elbows are used to change the direction of pipe flow, typically by 45 degrees or 90 degrees.
    - They are crucial for routing pipes around obstacles, corners, or tight spaces.
    - Elbows help in aligning pipes for easier installation and maintenance.
2. Reducer:
    - Reducers are used to connect pipes of different sizes, allowing for a smooth transition in diameter.
    - They ensure compatibility between pipes with different dimensions.
    - Reducers help in adapting the flow rate and pressure between two pipe sections.
3. Tee:
    - Tees are used for creating branching connections in a pipeline.

- They provide an outlet at a 90-degree angle to the main pipe, allowing for a separate line to be connected.
- Tees are commonly used in applications such as distributing water supply to multiple fixtures or creating parallel lines in a system.

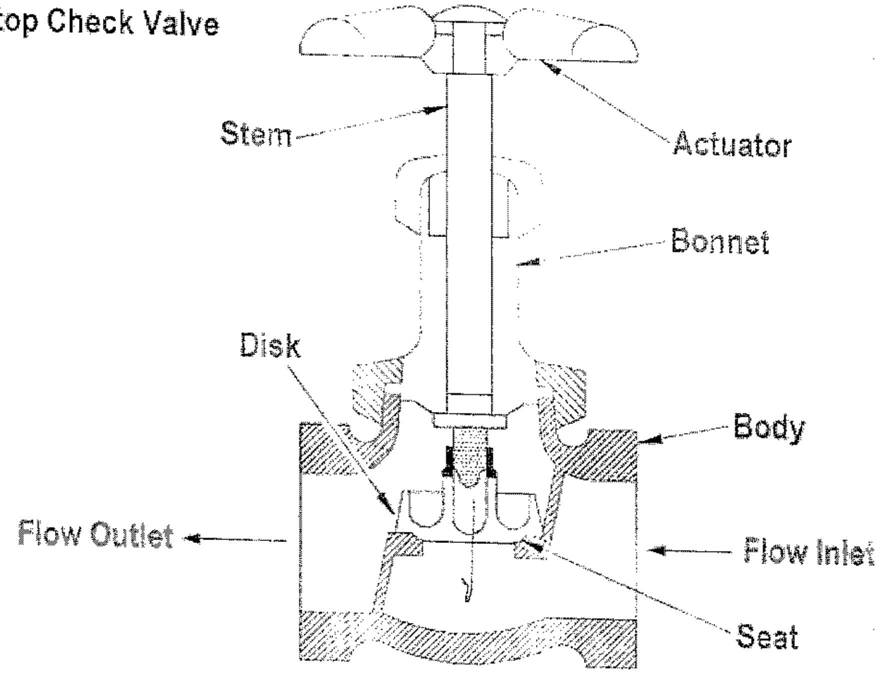
4. Coupling:
   - Couplings are used to join two pipes of the same size together.
   - They provide a secure and leak-resistant connection.
   - Couplings are commonly used for permanent installations where a strong and permanent joint is required.
5. Union:
   - Unions are used for easily disconnecting and reconnecting pipes without the need for cutting or disassembling the fittings.
   - They provide a convenient access point for maintenance or repairs.
   - Unions are commonly used in systems that may require periodic disconnection, such as pump connections or equipment servicing.
6. Cross:
   - Cross fittings allow branching in multiple directions, typically at right angles to each other.
   - They are used to create intersecting lines in complex pipe systems.
   - Cross fittings enable the distribution of fluid flow in various directions.
7. Flange:
   - Flanges are used to connect pipes by bolting them together.
   - They provide a strong and secure joint, particularly in high-pressure and large-diameter piping systems.
   - Flanges are commonly used in industries such as oil and gas, chemical processing, and power generation.
8. Cap:
   - Caps are used to seal the end of a pipe, blocking the flow.
   - They are typically used for capping off unused or terminated pipe ends.
   - Caps are useful for temporarily closing a pipe for maintenance or testing purposes.

VALVES FOR WATER WORKS

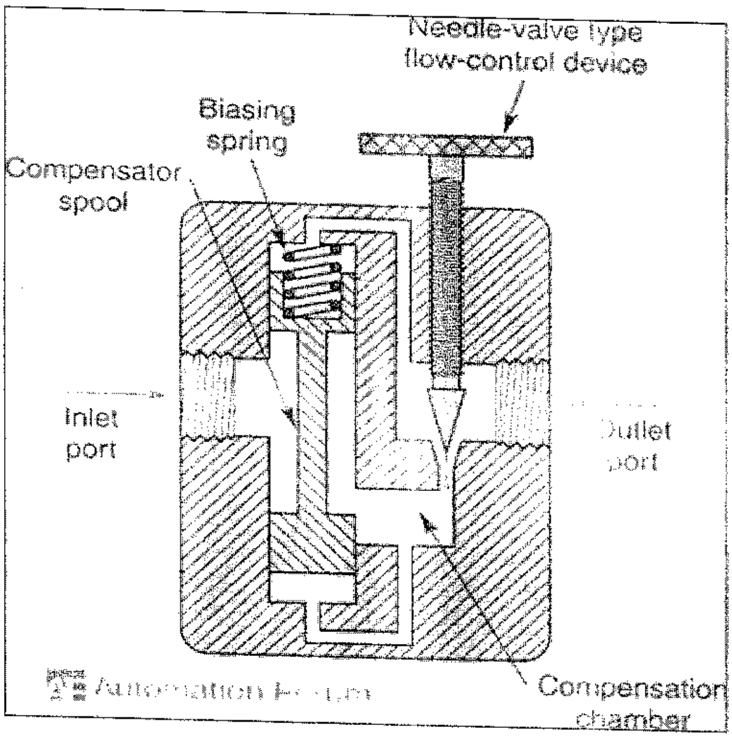
1. PRESSURE RELIEF VALVE DIAGRAM



2 Stop Check Valve

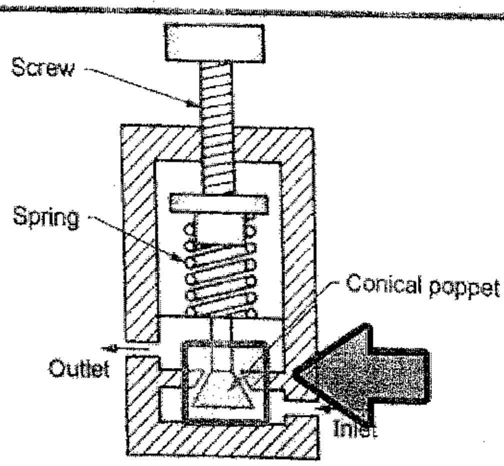


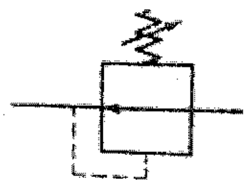
3 FLOW CONTROL VALVE DIAGRAM



4-

Pressure Reducing Valve





Symbol

These fittings. along with others, provide the necessary versatility, adaptability, and reliability in plumbing and piping systems, allowing for efficient and effective fluid transport and control. The specific choice of fittings depends on the system requirements, pipe dimensions, fluid characteristics, and industry standards.

Codes and Standards used in Pipe Fittings

Codes and **standards** play a crucial role in ensuring the quality, safety, and compatibility of pipe fittings used in various industries and applications. Here are some of the key codes and standards commonly used in the realm of pipe fittings:

1. **ASTM Standards**: The American Society for Testing and Materials (ASTM) develops and publishes standards for various materials, including metals, plastics, and rubber used in pipe fittings. These standards specify the material properties, dimensions, testing methods, and performance requirements for pipe fittings.
2. **ASME B16 Standards**: The American Society of Mechanical Engineers (ASME) publishes the B16 series of standards, which cover different types of pipe fittings. The most commonly referenced standards from this series include ASME B16.9 (Factory-Made Wrought Steel Butt-Welding Fittings), ASME B16.11 (Forged Fittings, Socket-Welding and Threaded), and ASME B16.5 (Pipe Flanges and Flanged Fittings).
3. **MSS Standards:** The Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) publishes standards related to pipe fittings. The MSS standards cover various aspects, including materials, dimensions, testing, and marking requirements for specific types of fittings, such as MSS SP-75 for high-strength, wrought, butt-welding fittings and MSS SP-97 for integrally reinforced forged branch outlet fittings.
4. **ANSI/ASME Standards**: The American National Standards Institute (ANSI) in conjunction with ASME establishes standards for pipe fittings. ANSI/ASME B31.1 (Power Piping) and ANSI/ASME B31.3 (Process Piping) provide guidelines for the design, fabrication, and installation of piping systems, including the use of pipe fittings.
5. **ISO Standards:** The International Organization for Standardization (ISO) develops international standards for various products. including pipe fittings. ISO standards provide specifications for

materials, dimensions, and performance characteristics of pipe fittings, ensuring global compatibility and interchangeability.

6. **API Standards**: The American Petroleum Institute (API) publishes standards specific to the oil and gas industry, including standards for pipe fittings used in pipelines and refineries. Examples of API standards related to pipe fittings include API 5L (Line Pipe). API 6D (Pipeline Valves), and API 600 (Steel Gate Valves).

These are just a few examples of the many codes and standards used in pipe fittings. It's important to consult the specific code or standard applicable to your industry, application, and geographical region to ensure compliance and proper selection of pipe fittings for your specific needs.

discharge excess water to prevent a potential explosion. Similarly, in a large water storage tank, the valve prevents damage caused by excessive pressure during filling or thermal expansion.

(b) Stop Valve - Gate Stop Valve:

Use and Applications:

- Stop valves, particularly gate stop valves, are used to control the flow of water and to isolate specific sections of a water supply system. They provide a full open or full closed position, making them suitable for applications where on/off control is needed.
- Practical Applications: Gate stop valves are commonly found in residential plumbing systems. They are used to shut off the water supply to individual fixtures, such as sinks, toilets, and showers. In commercial or industrial settings, they are used to isolate sections of the water supply for maintenance or emergency repairs.

(c) Flow Regulation Valve - Globe Valve:

Use and Applications:

- Flow regulation valves, like globe valves, are designed to provide precise control over the flow rate of water. They are used in applications where throttling and fine-tuned adjustments of flow are necessary.
- Practical Applications: Globe valves are commonly used in industrial settings, such as in water treatment plants, where accurate control of water flow is crucial. They are also used in residential plumbing systems for controlling the flow of water to showers, faucets, and appliances like washing machines.

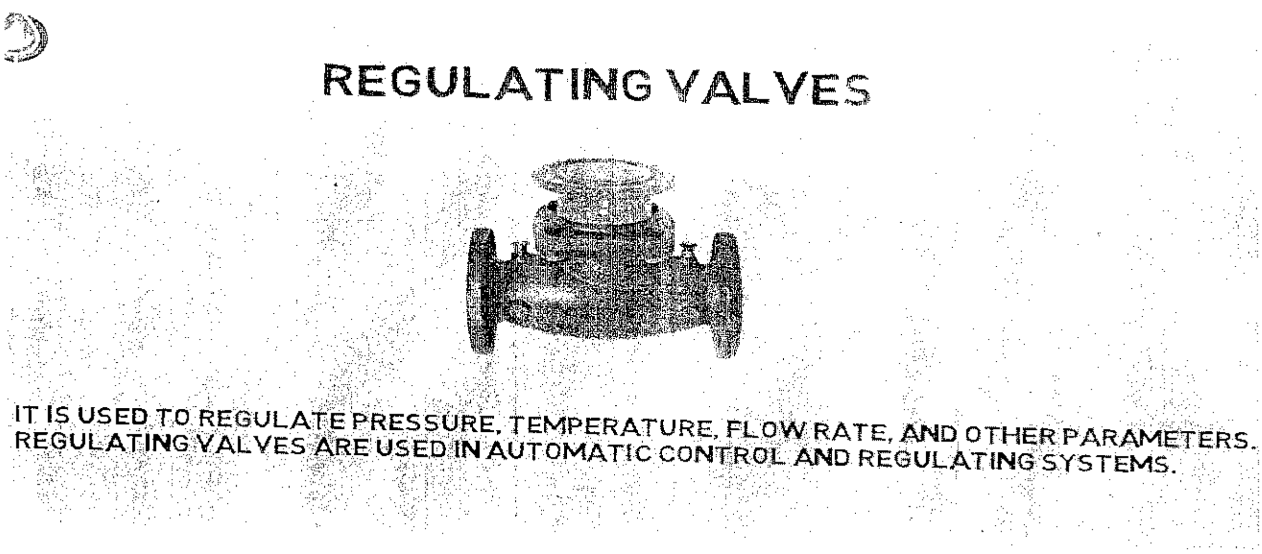
(d) Pressure Regulation Valve - Zurn Wilkins 70XLDU Pressure Reducing Valve:

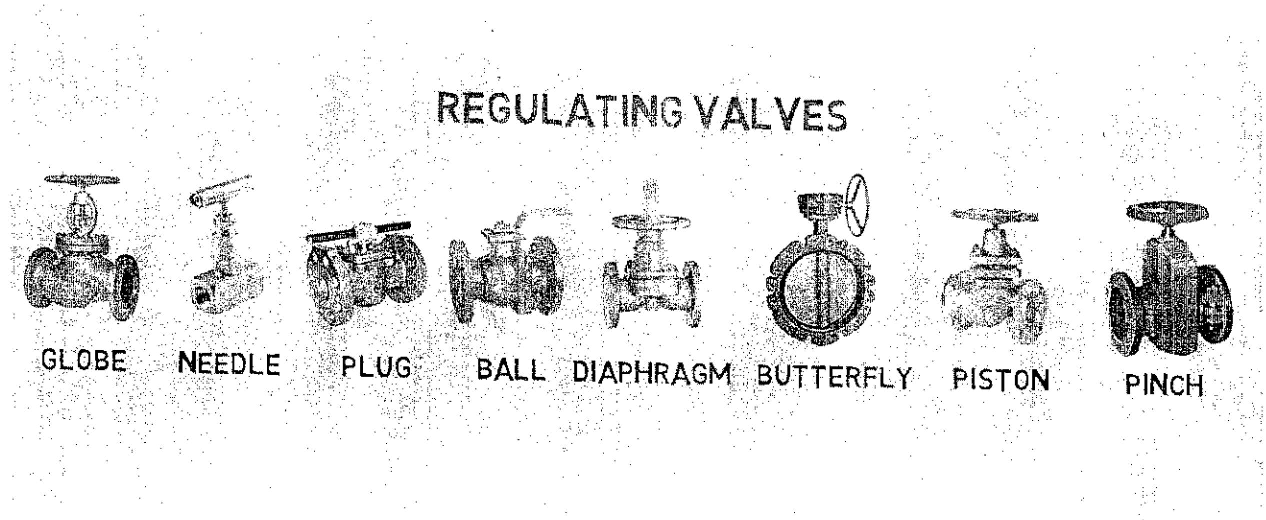
Use and Applications:

- Pressure regulation valves, specifically pressure reducing valves, are used to maintain a consistent and safe water pressure downstream from the main supply. They ensure that the pressure remains within an acceptable range for various applications.
- Practical Applications: Pressure reducing valves are widely used in residential, commercial, and industrial settings. In residential applications, they are installed to reduce the high pressure from the municipal supply to a safe level for domestic use. In commercial buildings and industries, pressure reducing valves are used to protect plumbing fixtures, appliances, and sensitive equipment that may be sensitive to high water pressures.

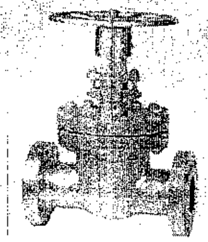
Overall, these valves play critical roles in maintaining the stability, safety, and efficiency of water supply systems in various practical applications. They are selected and installed based on the specific requirements of each system and the need for pressure regulation, flow control, or isolation.

PICTURES OFVARIOUS TYPES OF VALVES USED IN WATER WORKS



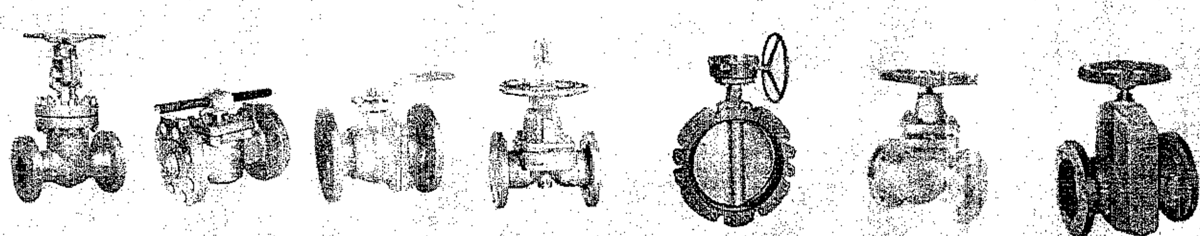


ISOLATING VALVES



THIS IS USED TO CONTROL THE FLOW. USUALLY FOR MAINTENANCE OR SAFETY PURPOSES.
IT IS INSTALLED WITHIN A PROPERTY IS REQUIRED TO BE IN AN ACCESSIBLE LOCATION.
THE LOCATION OF AN ISOLATING VALVE CAN VARY BASED ON THE TYPE AND LOCATION.

ISOLATING VALVES

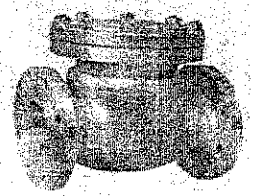


GATE PLUG BALL DIAPHRAGM BUTTERFLY PISTON PINCH

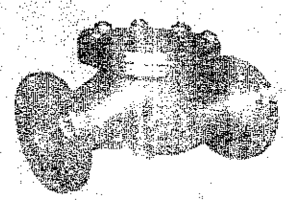
NON-RETURN VALVES

It prevents reversal of flow in the piping systems

NON-RETURN VALVES



SWING CHECK VALVE



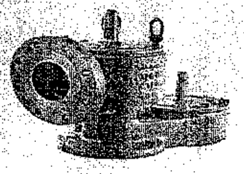
LIFT CHECK VALVE

Safety Relief Valve

A Relief valve used to control or limit the pressure in Piping s

Safety Relief Valve

Pressure Relief valve



Vaccume Relief valve

VALVES FOR SPECIAL PURPOSES

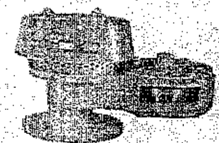
MULTI-PORT

FLUSH BOTTOM

FLOAT

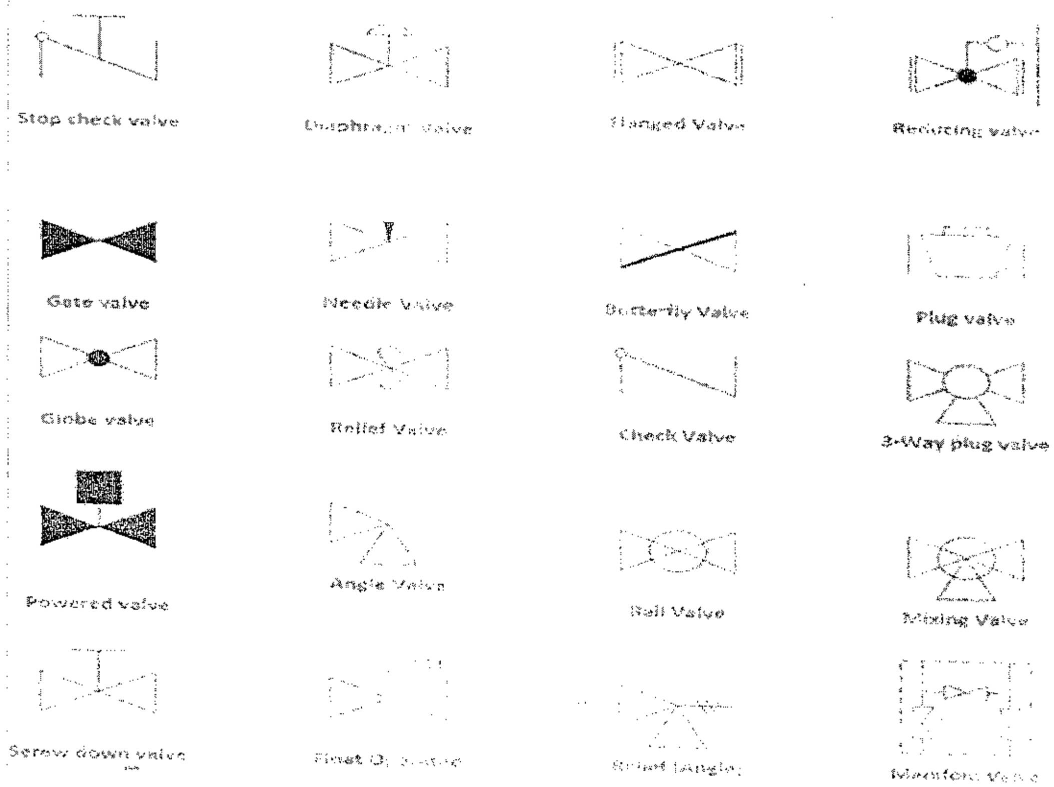
FOOT

PRESSURE RELIEF



BREATHER

REPRESENTATION OF VALVES ON DRAWINGS


Stop check valve
Flanged Valve
Gate valve
Needle Valve
Butterfly Valve
Plug valve
Globe valve
Relief Valve
Check Valve
3-Way plug valve
Powered valve
Angle Valve
Ball Valve
Mixing Valve
Screw down valve